

Exp: 01 Lexical Analyzer to identify identifiers, constants, operators

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
#include <string.h>
```

```
int isOperator(char ch) {  
    return (ch == '+' || ch == '-' || ch == '*' || ch == '/' ||  
            ch == '=' || ch == '%' || ch == '<' || ch == '>');  
}
```

```
int main() {  
    char input[500];  
    int i = 0;  
  
    printf("Enter an expression: ");  
    fgets(input, sizeof(input), stdin);  
  
    printf("\nTokens found:\n");
```

```
    while (input[i] != '\0') {  
        if (isspace((unsigned char)input[i])) {  
            i++;  
        }  
        else if (isalpha((unsigned char)input[i]) || input[i] == '_') {  
            char identifier[100];  
            int j = 0;  
  
            while (isalnum((unsigned char)input[i]) || input[i] == '_') {  
                identifier[j++] = input[i++];  
            }  
            identifier[j] = '\0';
```

```

        printf("Identifier: %s\n", identifier);
    }
    else if (isdigit((unsigned char)input[i])) {
        char constant[100];
        int j = 0;

        while (isdigit((unsigned char)input[i])) {
            constant[j++] = input[i++];
        }
        constant[j] = '\0';

        printf("Constant: %s\n", constant);
    }
    else if (isOperator(input[i])) {
        printf("Operator: %c\n", input[i]);
        i++;
    }
    else {
        i++;
    }
}

return 0;
}

```

Output:

Enter an expression: a=b+25*c

Tokens found:

Identifier: a

Operator: =

Identifier: b

Operator: +

Constant: 25

Operator: *

Identifier: c